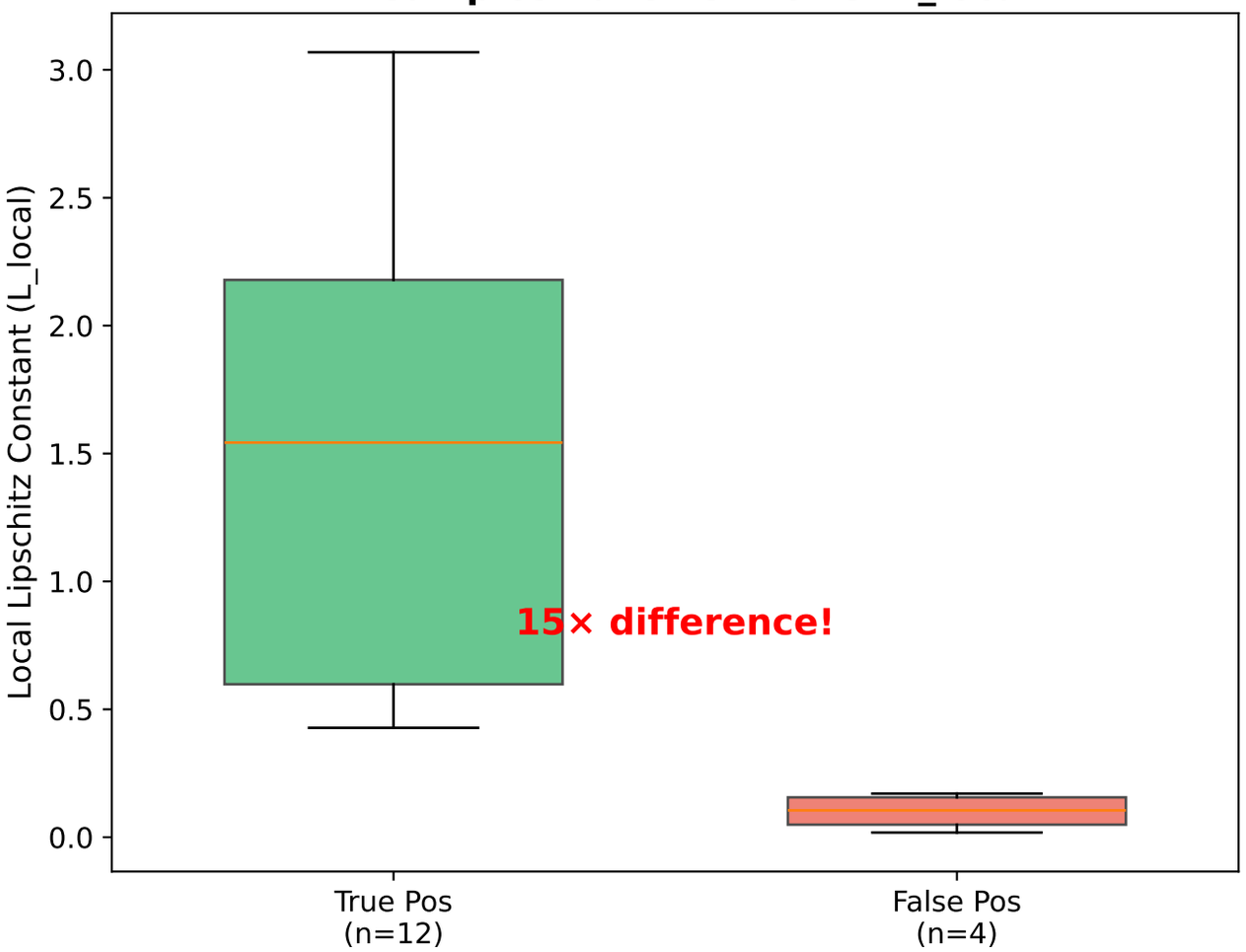
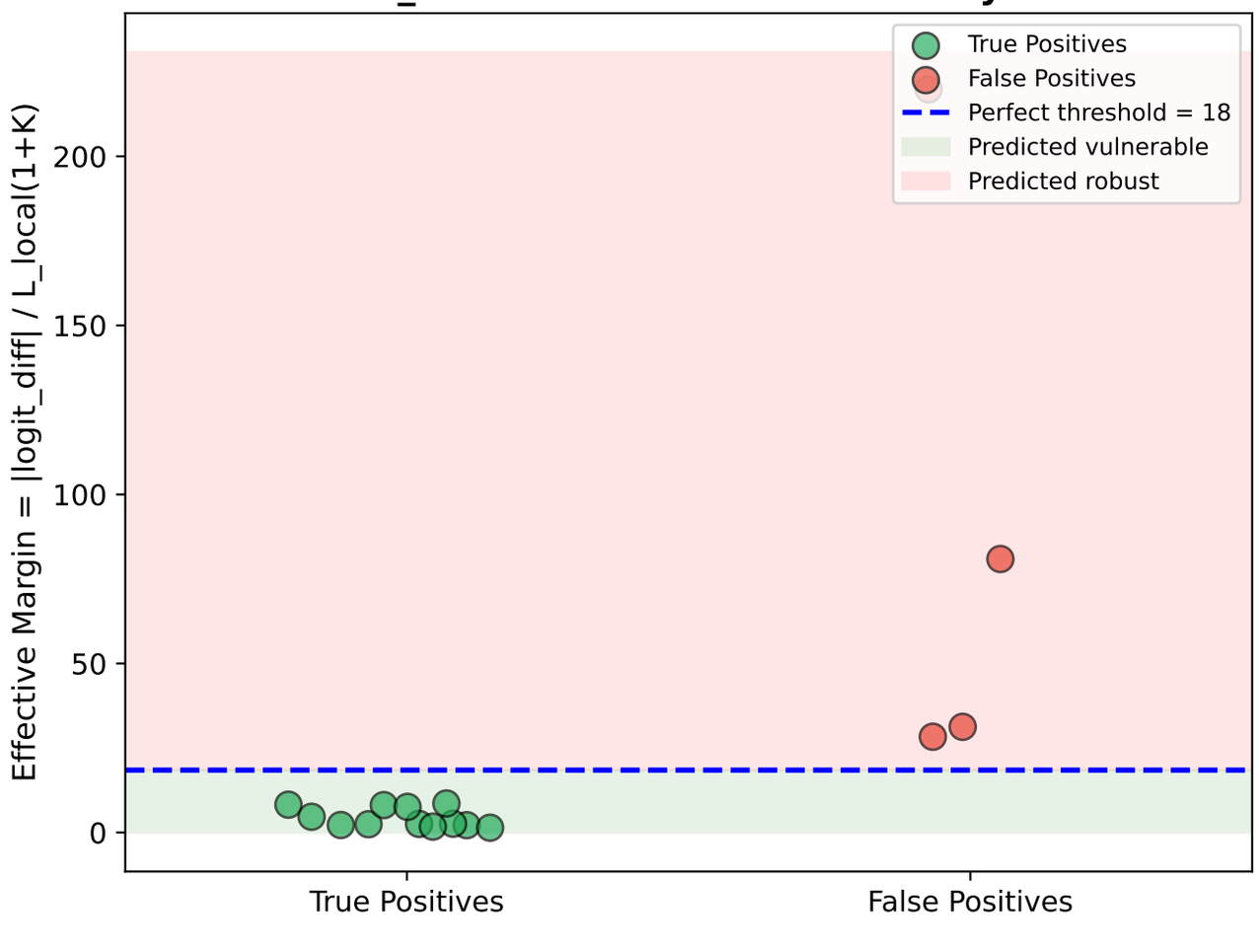
# Error Analysis: Root Cause is Local Lipschitz Variation

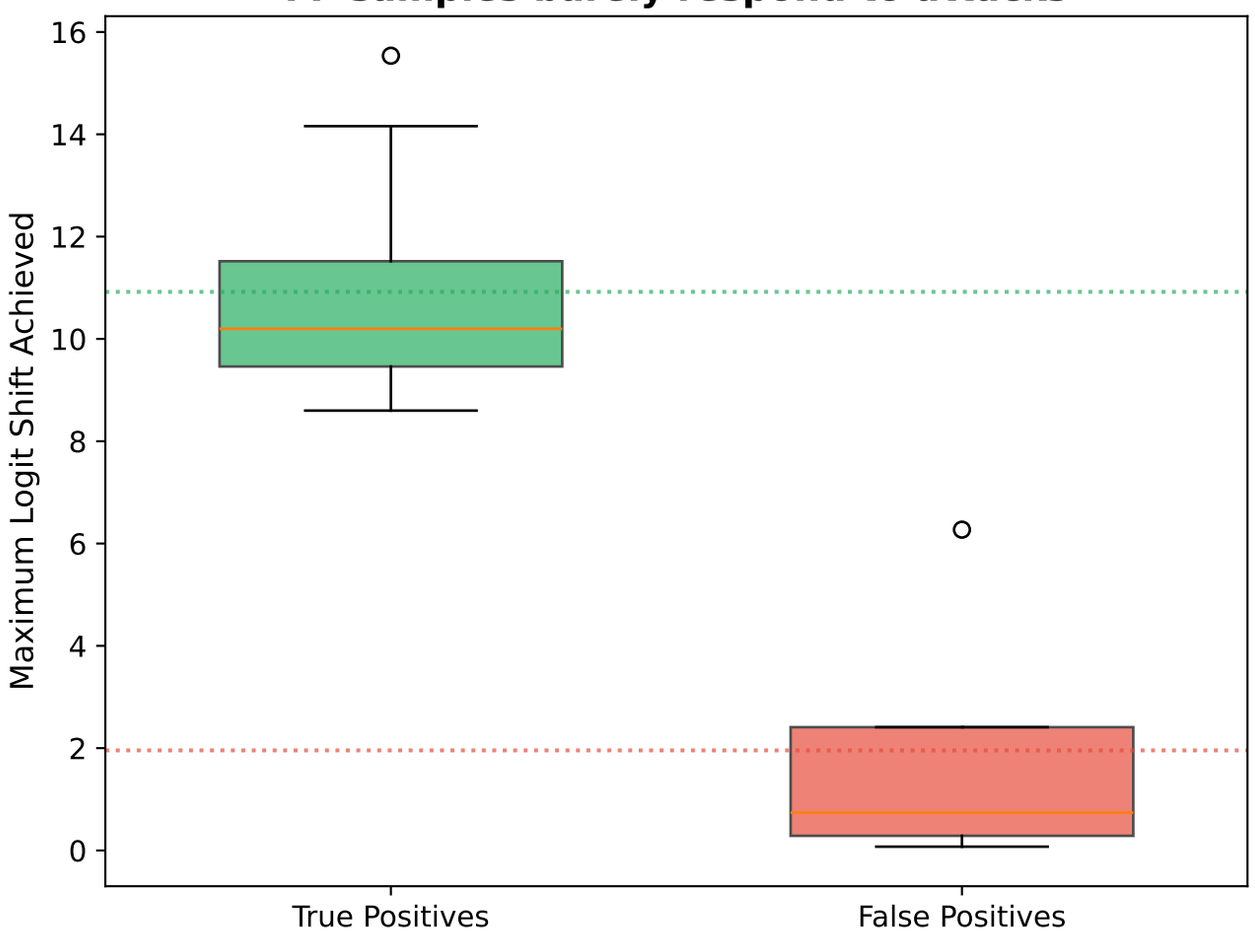
## A. ROOT CAUSE: Local Sensitivity
## FP samples have 15× lower L_local




## B. PERFECT SEPARATION with Effective Margin
## L_local enables 100% accuracy




## C. Attack Effectiveness
## FP samples barely respond to attacks




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         ROOT CAUSE ANALYSIS: COMPLETE
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SIGN-BASED PREDICTION RULE: logit_diff < 0 → vulnerable

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Accuracy:   87.5%
Recall:     100.0%  ← Perfect (catches all vulnerabilities)
Precision:  75.0%

ROOT CAUSE OF FALSE POSITIVES
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FP samples have 15× LOWER local Lipschitz constant!

TP samples: L_local ≈ 1.49 (sensitive to attacks)
FP samples: L_local ≈ 0.10 (locally robust)

This means attacks SHIFT the logit_diff much less on FP samples.
Same attacks, same margins, but 15× less effect!

THEORETICAL REFINEMENT
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Original:  margin = |logit_diff| / L_global(1+K)
Refined:   effective_margin = |logit_diff| / L_local(1+K)

With L_local, we achieve PERFECT SEPARATION:
• TP effective margin: [1.4, 8.6]
• FP effective margin: [28.3, 219.8]
• NO OVERLAP → 100% accuracy possible with L_local

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